Listen

Week 12 Lab: Inheritance, Polymorphism & Special Methods

What Are We Building?

Last week you built classes. This week you'll learn to make classes **inherit** from each other — so a child class can reuse everything from a parent and add its own behavior. You'll also learn **special methods** (dunder methods) that let your objects work with `print()`, `len()`, `==`, and `sorted()`.

Open the starter files and look for the **`# TODO`** markers.

Q1: Scanner Inheritance — Parent and Child Classes

File: `lab12_q1_starter.py`

What if you need different scanners — one for ports, another for HTTP pages? Instead of two separate classes, create a **parent class** with shared code, then **child classes** that add specific behavior.

The starter has the parent class structure and main section. **Your work is completing the parent method and building two child classes.**

`Scanner` (parent) — shared `display_results()` method

`PortScanner(Scanner)` — scans ports using `socket.connect_ex()`

`HTTPScanner(Scanner)` — checks HTTP paths using `urllib`

Reference

```python
# Inheritance — child class gets everything from parent
class Animal:
    def __init__(self, name):
        self.name = name

class Dog(Animal):
    def __init__(self, name, breed):
        super().__init__(name)         # call parent's __init__
        self.breed = breed

d = Dog("Rex", "Lab")
print(d.name)      # "Rex" — inherited from Animal
```

Test It

Run `python lab12_q1_starter.py`. You should see port scan results and HTTP path check results.

Q2: Dunder Methods — Making Objects Smarter

File: [lab12_q2_starter.py]

You'll add special methods to `Finding` and `Report` so they can be compared (`==`), sorted (`sorted()`), counted (`len()`), and merged (`+`).

The starter has the class structure and main section. **Your work is 4 dunder methods.**

`Finding` — add `__eq__` and `__lt__`

`Report` — add `__len__` and `__add__`

Reference

```python
class Box:
    def __init__(self, size):
        self.size = size
    def __eq__(self, other):     # ==
        return self.size == other.size
    def __lt__(self, other):     # < (enables sorted())
```

```
        return self.size < other.size
    def __len__(self):          # len()
        return self.size
    def __add__(self, other):   # +
        return Box(self.size + other.size)
```

Test It

Run `python lab12_q2_starter.py`. You should see findings compared, sorted LOW→HIGH, reports counted, and two reports merged.

Submission

- Only submit your GitHub (No Screenshot, no ZIP or Python file, just the Link) repository link, for example: **https://github.com/sojoudian/COMP2152_Labs**
- Create a branch named **`lab_week12`**, complete your lab work in that branch, and once finished, merge it into your **`master`** branch.